Discovery Pattern Routing for `p-vs-np`

Neuro-symbolic AI Group
Idiap Research Institute

March 18, 2026

Abstract

This report turns blocked-frontier prompt packs into an executable discovery-routing layer. It spawns dedicated discovery layers per breakthrough family and per cross-cutting epistemic pattern, then orders them to avoid collapsing into one depth-first search line.

1 Summary

- Discovery layers are now spawned from prompt packs rather than stored only as passive prompt artifacts.
- Routing alternates breakthrough-family branches with cross-cutting epistemic-pattern branches to avoid indeterminate depth-first search.
- Each layer records its own theorem target, stop criteria, and expected deliverables so discovery remains inspectable and finite.
- Each layer also carries a local-first heuristic for when to search externally for a mathematical framework with similar structural properties.

2 Routing Policy

- Prioritize pressure on the blocked frontier, but enforce diversity across layer kinds and breakthrough families.
- Use family-specific layers to attack concrete bridge defects and epistemic-pattern layers to change representation, invariants, or decomposition when local search stalls.
- Escalate or stop when a layer only reproduces its anti-patterns or fails to yield a theorem-shaped claim.
- Trigger external framework search only after a recorded local failure, and only when the missing move is better modeled as a borrowed foundation than as another local lemma.

3 Dispatch Rounds

Round 1

Rationale. Alternate targeted family branches with broader epistemic patterns to avoid collapsing into one depth-first line of search.

Layers. disclayer.p-vs-np.pattern.invariant-discovery, disclayer.p-vs-np.family.abstraction-search, disclayer.p-vs-np.pattern.regime-localization

Covered breakthrough families. abstraction-search, realization, transport, bridge-specificity, barrier-aware transfer, alignment, explicitness, constructive instantiation, semantic bridge

Round 2

Rationale. Alternate targeted family branches with broader epistemic patterns to avoid collapsing into one depth-first line of search.

Layers. disclayer.p-vs-np.family.realization, disclayer.p-vs-np.pattern.hybridization-of-partial-bridges, disclayer.p-vs-np.family.bridge-specificity

Covered breakthrough families. realization, abstraction-search, transport, bridge-specificity, explicitness, constructive instantiation, semantic bridge

Round 3

Rationale. Alternate targeted family branches with broader epistemic patterns to avoid collapsing into one depth-first line of search.

Layers. disclayer.p-vs-np.pattern.counterexample-led-concept-formation, disclayer.p-vs-np.family.transport, disclayer.p-vs-np.pattern.route-decomposition

Covered breakthrough families. abstraction-search, realization, transport, barrier-aware transfer, alignment, semantic bridge, bridge-specificity

Round 4

Rationale. Alternate targeted family branches with broader epistemic patterns to avoid collapsing into one depth-first line of search.

Layers. disclayer.p-vs-np.family.barrier-aware-transfer, disclayer.p-vs-np.pattern.problem-representation-shift, disclayer.p-vs-np.family.alignment

Covered breakthrough families. barrier-aware transfer, abstraction-search, transport, alignment, semantic bridge

Round 5

Rationale. Alternate targeted family branches with broader epistemic patterns to avoid collapsing into one depth-first line of search.

Layers. disclayer.p-vs-np.family.explicitness, disclayer.p-vs-np.family.constructive-instantiation, disclayer.p-vs-np.family.semantic-bridge

Covered breakthrough families. explicitness, constructive instantiation, semantic bridge

Round 6

Rationale. Alternate targeted family branches with broader epistemic patterns to avoid collapsing into one depth-first line of search.

Layers. disclayer.p-vs-np.family.categorical-reframing, disclayer.p-vs-np.family.equivalence-coherence-reframing

Covered breakthrough families. realization, transport, bridge-specificity, barrier-aware transfer, alignment, constructive instantiation, semantic bridge, explicitness

4 disclayer.p-vs-np.pattern.invariant-discovery

Kind. epistemic-pattern

Target. invariant-discovery

Routing priority. 6.868

Dispatch rank. 1

Spawn reason. Spawn 'invariant-discovery' as a cross-cutting discovery layer because the blocked frontier spans multiple breakthrough families and needs a broader reframing.

Rationale. Several live routes are still searching for the right preserved quantity or witness measure, so invariant discovery remains a first-class discovery move.

System Prompt

You are searching for the invariant that the current route is really trying to preserve, expose, or measure. Do not begin from the named bridge object. Begin from the quantity, structure, or monotone property that would survive the route and explain the payoff.

User Task

Identify 2-4 candidate invariants, state how each is computed or witnessed, and say which one would most sharply reorganize the current blocked route.

Expected Deliverables

- Produce one reframing under 'invariant-discovery' that changes the search state rather than restating the blocked lemma.
- Connect the reframing back to a theorem-shaped objective compatible with 'There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.'.
- Record one stop condition showing when the pattern has become superficial rather than discovery-bearing.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'accepting existence without algorithmic access'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'invariant-discovery' only re-labels the blocked frontier instead of reducing proof debt or changing the representation.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. structural-analogy, cross-family-pressure, pattern-level-reframing, bridge-family-foundation-gap

Candidate framework regions. potential / energy method frameworks, monotone measure frameworks, certificate-complexity and witness-measure theories, proof complexity, pseudorandomness / derandomization

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Invariant Discovery Prompt with preserved invariant and bridge theorem
- Invariant Discovery Prompt correspondence theorem transport invariant
- Invariant Discovery Prompt framework Several live routes are still searching for the right preserved quantity or witness measure, so invariant discovery remains a first-class discovery move. theorem
- theorem or framework matching: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.

Required Questions

- What quantity or structure must survive every legitimate route variant?
- Which candidate invariant is strong enough to force the payoff but weak enough to be realizable?
- How would failure of the invariant candidate sharpen the obstruction class?

Search Modes

- activation theorem search
- barrier-aware gadget design

- bridge split
- bridge-localization refinement
- canonical example construction
- certificate extraction
- certificate map construction
- counterfactual sharpening
- cross-model transport principle
- dual characterization of the target object
- duality or normal form search
- effective construction
- equivalence search between live carrier encodings
- example-preserving reduction search
- explicit carrier construction
- explicit extractor or generator search
- generalization by faithful embedding
- hypothesis verification search
- impossibility theorem search
- invariant-preserving compilation theorem
- normal-form mediation
- object-class correspondence search
- parameter-preserving reformulation
- problem-specific to generic theorem bridge
- reduction regime decomposition
- representation normal form for formula-size transfer
- semantic correspondence theorem
- semantic-to-algorithmic upgrade theorem
- surrogate problem design with forcing semantics
- template instantiation
- two-way implication decomposition
- typed abstraction search
- uniformization

5 disclayer.p-vs-np.family.abstraction-search

Kind. breakthrough-family

Target. abstraction-search

Routing priority. 6.443

Dispatch rank. 2

Spawn reason. Spawn a dedicated 'abstraction-search' discovery layer because obstruction 'mcsp-magnification' carries repeated pressure across the blocked frontier.

Rationale. Find a sharper abstraction that turns the blocked route into a typed, inspectable theorem target.

System Prompt

You are searching for the minimal conceptual breakthrough that would turn a blocked route into a typed theorem target. Do not restate the blocked lemma. Identify the sharper carrier, invariant, or bridge law that would make the next theorem materially more specific.

User Task

Restate the obstruction as a missing abstraction problem, propose 2-4 typed bridge candidates, and state the first theorem-pair-specific lemma each candidate would make possible.

Expected Deliverables

- State one theorem-shaped candidate centered on 'abstraction-search' satisfying 'State a route-specific theorem that would resolve the blocked interface and make the next proof obligation materially sharper.'.
- Make 'A theorem-family-specific carrier, invariant, or bridge law that adds new mathematical content.' explicit enough that an expert could inspect the object or transfer law.
- Record one decisive negative route explaining what would fail if this discovery layer does not land.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'repeating the same blocked statement without new typed structure'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'abstraction-search' remains purely schematic after the first explicit object or theorem candidate.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. cross-family-pressure, bridge-family-foundation-gap

Candidate framework regions. proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Abstraction Search Prompt with preserved invariant and bridge theorem
- Abstraction Search Prompt correspondence theorem transport invariant
- Abstraction Search Prompt framework Find a sharper abstraction that turns the blocked route into a typed, inspectable theorem target. theorem
- theorem or framework matching: State a route-specific theorem that would resolve the blocked interface and make the next proof obligation materially sharper.

Required Questions

- What typed object would add mathematical content beyond the current schematic carrier?
- What route-specific theorem would become stateable if that object existed?
- Which candidate clarifies the negative route if it fails?

Search Modes

- typed abstraction search
- bridge split
- counterfactual sharpening

6 disclayer.p-vs-np.pattern.regime-localization

Kind. epistemic-pattern

Target. regime-localization

Routing priority. 6.758

Dispatch rank. 3

Spawn reason. Spawn 'regime-localization' as a cross-cutting discovery layer because the blocked frontier spans multiple breakthrough families and needs a broader reframing.

Rationale. The frontier repeatedly crosses model or threshold boundaries, so discovery should search for the smallest regime in which the route first becomes true.

System Prompt

You are searching for the smallest model, threshold, asymptotic regime, or promise-gap setting in which the route might first become true. Do not ask for the full theorem immediately if a more local regime would already be a major contribution.

User Task

Localize the frontier by proposing restricted regimes, threshold windows, or case splits where the target theorem may first activate, and explain what each localized success would imply for the broader programme.

Expected Deliverables

- Produce one reframing under 'regime-localization' that changes the search state rather than restating the blocked lemma.
- Connect the reframing back to a theorem-shaped objective compatible with 'There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.'.
- Record one stop condition showing when the pattern has become superficial rather than discovery-bearing.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'adding more obligations without exhibiting a concrete carrier family'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'regime-localization' only re-labels the blocked frontier instead of reducing proof debt or changing the representation.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. structural-analogy, cross-family-pressure, pattern-level-reframing, bridge-family-foundation-gap

Candidate framework regions. threshold and regime-restriction frameworks, restricted-model lower-bound programmes, proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Regime Localization Prompt with preserved invariant and bridge theorem
- Regime Localization Prompt correspondence theorem transport invariant
- Regime Localization Prompt framework The frontier repeatedly crosses model or threshold boundaries, so discovery should search for the smallest regime in which the route first becomes true. theorem
- theorem or framework matching: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.

Required Questions

- Which regime change most plausibly removes the current obstruction without trivializing the result?
- What is the first nontrivial threshold at which the route may become true?
- Would a localized theorem already count as an independently valuable subresult?

Search Modes

- activation theorem search
- barrier-aware gadget design
- bridge split
- bridge-localization refinement
- canonical example construction

- counterfactual sharpening
- cross-model transport principle
- equivalence search between live carrier encodings
- example-preserving reduction search
- explicit carrier construction
- generalization by faithful embedding
- hypothesis verification search
- impossibility theorem search
- invariant-preserving compilation theorem
- parameter-preserving reformulation
- problem-specific to generic theorem bridge
- reduction regime decomposition
- representation normal form for formula-size transfer
- surrogate problem design with forcing semantics
- template instantiation
- typed abstraction search

7 disclayer.p-vs-np.family.realization

Kind. breakthrough-family

Target. realization

Routing priority. 6.443

Dispatch rank. 4

Spawn reason. Spawn a dedicated 'realization' discovery layer because obstruction 'carrier-realization' carries repeated pressure across the blocked frontier.

Rationale. Turn a typed but schematic carrier into an explicit mathematical object family with admissibility conditions and a forcing theorem.

System Prompt

You are converting a typed but schematic carrier into an explicit mathematical object family. Do not rename the carrier. Produce admissibility conditions, a realizability route from the source theorem, and the forcing lemma needed at the target.

User Task

Construct or sharply constrain an explicit witness family, obstruction family, surrogate promise problem, or quantitative measure that realizes the blocked route from source theorem to target theorem.

Expected Deliverables

- State one theorem-shaped candidate centered on 'realization' satisfying 'If source theorem S holds in regime R, then one can construct an explicit carrier C with admissibility conditions A such that C preserves invariant I and supports a route-specific forcing lemma to target T.'.
- Make 'An explicit obstruction family, witness family, surrogate promise problem, or quantitative measure that is realized from the source theorem rather than only named by role.' explicit enough that an expert could inspect the object or transfer law.
- Record one decisive negative route explaining what would fail if this discovery layer does not land.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'renaming the carrier without adding realizability conditions'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'realization' remains purely schematic after the first explicit object or theorem candidate.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. structural-analogy, cross-family-pressure, bridge-family-foundation-gap

Candidate framework regions. proof complexity witness measures, coding theory, pseudorandom objects and hardness amplification

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Carrier Realization Prompt with preserved invariant and bridge theorem
- Carrier Realization Prompt correspondence theorem transport invariant

- Carrier Realization Prompt framework Turn a typed but schematic carrier into an explicit mathematical object family with admissibility conditions and a forcing theorem. theorem
- theorem or framework matching: If source theorem S holds in regime R, then one can construct an explicit carrier C with admissibility conditions A such that C preserves invariant I and supports a route-specific forcing lemma to target T.

Required Questions

- What is the concrete carrier set, map, family, or measure?
- How is it built from the source theorem rather than postulated?
- What admissibility conditions prevent it from being only a relabeling?

Search Modes

- explicit carrier construction
- equivalence search between live carrier encodings
- surrogate problem design with forcing semantics

8 disclayer.p-vs-np.pattern.hybridization-of-partial-bridges

Kind. epistemic-pattern

Target. hybridization-of-partial-bridges

Routing priority. 6.728

Dispatch rank. 5

Spawn reason. Spawn 'hybridization-of-partial-bridges' as a cross-cutting discovery layer because the blocked frontier spans multiple breakthrough families and needs a broader reframing.

Rationale. Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored.

System Prompt

Do not assume one bridge object must do all the work. Search for a hybrid route in which two partial bridges compose, with each bridge carrying a distinct inferential burden.

User Task

Identify complementary partial bridges, specify their composition order, and state the intermediate theorem that would let them work together.

Expected Deliverables

- Produce one reframing under 'hybridization-of-partial-bridges' that changes the search state rather than restating the blocked lemma.
- Connect the reframing back to a theorem-shaped objective compatible with 'There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem

Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.'.

- Record one stop condition showing when the pattern has become superficial rather than discovery-bearing.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'accepting existence without algorithmic access'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'hybridization-of-partial-bridges' only re-labels the blocked frontier instead of reducing proof debt or changing the representation.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. structural-analogy, cross-family-pressure, pattern-level-reframing, bridge-family-foundation-gap

Candidate framework regions. composed transfer frameworks, hybrid argument methods, multi-interface reduction theories, proof complexity, pseudorandomness / derandomization

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Hybrid Bridge Synthesis Prompt with preserved invariant and bridge theorem
- Hybrid Bridge Synthesis Prompt correspondence theorem transport invariant
- Hybrid Bridge Synthesis Prompt framework Some blocked routes sit at the intersection of multiple breakthrough families, which is evidence that one bridge object may be insufficient and a composed route should be explored. theorem

- theorem or framework matching: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.

Required Questions

- Which blocked routes solve different parts of the same theorem pair?
- What intermediate object or theorem would allow the partial bridges to compose?
- What failure would show that the hybrid route is illegitimate rather than merely incomplete?

Search Modes

- activation theorem search
- bridge split
- bridge-localization refinement
- canonical example construction
- certificate extraction
- certificate map construction
- counterfactual sharpening
- cross-model transport principle
- duality or normal form search
- effective construction
- equivalence search between live carrier encodings
- example-preserving reduction search
- explicit carrier construction
- explicit extractor or generator search
- hypothesis verification search
- invariant-preserving compilation theorem
- representation normal form for formula-size transfer
- semantic correspondence theorem
- semantic-to-algorithmic upgrade theorem
- surrogate problem design with forcing semantics
- template instantiation
- two-way implication decomposition

- typed abstraction search
- uniformization

9 disclayer.p-vs-np.family.bridge-specificity

Kind. breakthrough-family

Target. bridge-specificity

Routing priority. 5.669

Dispatch rank. 6

Spawn reason. Spawn a dedicated 'bridge-specificity' discovery layer because obstruction 'target-grounding' carries repeated pressure across the blocked frontier.

Rationale. Stop treating the target theorem as the open problem and relocate search to the bridge that would activate it.

System Prompt

The target theorem is already known. Your task is to identify the local activation theorem that would verify its hypotheses from current artifacts. Do not rediscover the target theorem.

User Task

State the activation theorem under search, name the exact target hypothesis that is still unverified, and identify the local bridge object or transfer law needed to verify it.

Expected Deliverables

- State one theorem-shaped candidate centered on 'bridge-specificity' satisfying 'Target theorem T is fixed. The real theorem under search is the local activation theorem A that verifies T's hypotheses from current artifacts.'.
- Make 'No new target theorem; discovery should instead seek the carrier, transfer law, or alignment theorem that makes the target theorem applicable.' explicit enough that an expert could inspect the object or transfer law.
- Record one decisive negative route explaining what would fail if this discovery layer does not land.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'rediscovering the target theorem'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'bridge-specificity' remains purely schematic after the first explicit object or theorem candidate.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. cross-family-pressure

Candidate framework regions. activation-theorem frameworks, hypothesis-verification templates from neighboring areas

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Activation Theorem Prompt with preserved invariant and bridge theorem
- Activation Theorem Prompt correspondence theorem transport invariant
- Activation Theorem Prompt framework Stop treating the target theorem as the open problem and relocate search to the bridge that would activate it. theorem
- theorem or framework matching: Target theorem T is fixed. The real theorem under search is the local activation theorem A that verifies T's hypotheses from current artifacts.

Required Questions

- Which target-theorem hypothesis is still open locally?
- What bridge-local theorem would certify that hypothesis?
- What novelty remains once the target theorem itself is removed from search?

Search Modes

- activation theorem search
- hypothesis verification search
- bridge-localization refinement

10 disclayer.p-vs-np.pattern.counterexample-led-concept-formation

Kind. epistemic-pattern

Target. counterexample-led-concept-formation

Routing priority. 6.658

Dispatch rank. 7

Spawn reason. Spawn 'counterexample-led-concept-formation' as a cross-cutting discovery layer because the blocked frontier spans multiple breakthrough families and needs a broader reframing.

Rationale. The frontier is heavily blocked, so near-counterexamples and barrier analogs should be used to induce better concepts rather than only to reject claims.

System Prompt

Use failure as a discovery tool. Search for near-counterexamples, barrier analogs, and route-breaking examples, then ask what concept would separate the surviving cases from the failures.

User Task

Identify the strongest nearby failure patterns, describe what they invalidate, and derive a refined concept or admissibility condition from those failures.

Expected Deliverables

- Produce one reframing under 'counterexample-led-concept-formation' that changes the search state rather than restating the blocked lemma.
- Connect the reframing back to a theorem-shaped objective compatible with 'There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.'.
- Record one stop condition showing when the pattern has become superficial rather than discovery-bearing.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'adding a stronger carrier without specifying the target object class map'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'counterexample-led-concept-formation' only re-labels the blocked frontier instead of reducing proof debt or changing the representation.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. structural-analogy, cross-family-pressure, bridge-family-foundation-gap

Candidate framework regions. barrier and impossibility frameworks, minimal obstruction and forbidden-configuration theories, proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Counterexample-Led Concept Formation Prompt with preserved invariant and bridge theorem
- Counterexample-Led Concept Formation Prompt correspondence theorem transport invariant
- Counterexample-Led Concept Formation Prompt framework The frontier is heavily blocked, so near-counterexamples and barrier analogs should be used to induce better concepts rather than only to reject claims. theorem
- theorem or framework matching: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.

Required Questions

- What is the nearest counterexample or barrier analog to the current route?
- Which current concept fails to distinguish the positive and negative cases?
- What new concept becomes visible only after inspecting the failure pattern?

Search Modes

- barrier-aware gadget design
- bridge split

- counterfactual sharpening
- cross-model transport principle
- dual characterization of the target object
- duality or normal form search
- equivalence search between live carrier encodings
- explicit carrier construction
- generalization by faithful embedding
- impossibility theorem search
- invariant-preserving compilation theorem
- normal-form mediation
- object-class correspondence search
- parameter-preserving reformulation
- problem-specific to generic theorem bridge
- reduction regime decomposition
- representation normal form for formula-size transfer
- semantic correspondence theorem
- surrogate problem design with forcing semantics
- two-way implication decomposition
- typed abstraction search

11 disclayer.p-vs-np.family.transport

Kind. breakthrough-family

Target. transport

Routing priority. 5.610

Dispatch rank. 8

Spawn reason. Spawn a dedicated 'transport' discovery layer because obstruction 'formula-size-transport' carries repeated pressure across the blocked frontier.

Rationale. Find a theorem that preserves the relevant invariant across a cross-model move into formula-size.

System Prompt

You are searching for an invariant-preserving transport theorem across a model shift. Do not strengthen the source theorem unless it changes the transport mechanism. Focus on the cross-model map and the preserved quantitative parameter.

User Task

State the strongest plausible transport law, compilation theorem, or normal form that could move the current carrier into the target model without losing the invariant needed for the payoff.

Expected Deliverables

- State one theorem-shaped candidate centered on 'transport' satisfying 'For every admissible carrier C in the nonuniform-circuit regime, there is a transformed carrier C' in the formula-size regime preserving invariant I and retaining enough quantitative strength to imply target T.'.
- Make 'A transport law, compilation theorem, or complexity measure that is stable under the source-to-target model shift.' explicit enough that an expert could inspect the object or transfer law.
- Record one decisive negative route explaining what would fail if this discovery layer does not land.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'assuming the target theorem already provides the transport'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'transport' remains purely schematic after the first explicit object or theorem candidate.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. structural-analogy, cross-family-pressure, bridge-family-foundation-gap

Candidate framework regions. formula lower-bound transfer frameworks, communication complexity model-transfer methods, representation/normal-form theorems in circuit complexity

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Transport Principle Prompt with preserved invariant and bridge theorem
- Transport Principle Prompt correspondence theorem transport invariant
- Transport Principle Prompt framework Find a theorem that preserves the relevant invariant across a cross-model move into formula-size. theorem
- theorem or framework matching: For every admissible carrier C in the nonuniform-circuit regime, there is a transformed carrier C' in the formula-size regime preserving invariant I and retaining enough quantitative strength to imply target T.

Required Questions

- What exact quantity must be preserved across the model shift?
- What transformed carrier should live in the target model?
- What is the first obstruction showing why naive transport fails?

Search Modes

- cross-model transport principle
- invariant-preserving compilation theorem
- representation normal form for formula-size transfer

12 disclayer.p-vs-np.pattern.route-decomposition

Kind. epistemic-pattern

Target. route-decomposition

Routing priority. 6.578

Dispatch rank. 9

Spawn reason. Spawn 'route-decomposition' as a cross-cutting discovery layer because the blocked frontier spans multiple breakthrough families and needs a broader reframing.

Rationale. High-pressure obstruction classes are carrying multiple inferential roles at once, so route decomposition is needed before more local search.

System Prompt

The current route may be carrying too many inferential roles at once. Decompose it into smaller interfaces, each with its own object, invariant, and theorem target.

User Task

Split the blocked route into a sequence of subinterfaces, identify the minimal theorem at each interface, and state which subresult would already be valuable if the full route remained open.

Expected Deliverables

- Produce one reframing under 'route-decomposition' that changes the search state rather than restating the blocked lemma.
- Connect the reframing back to a theorem-shaped objective compatible with 'There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.'.
- Record one stop condition showing when the pattern has become superficial rather than discovery-bearing.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'adding more obligations without exhibiting a concrete carrier family'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'route-decomposition' only re-labels the blocked frontier instead of reducing proof debt or changing the representation.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. structural-analogy, cross-family-pressure, pattern-level-reframing, bridge-family-foundation-gap

Candidate framework regions. interface decomposition frameworks, multi-stage lifting and factorization methods, proof complexity, pseudorandomness / derandomization, coding-theoretic hardness surrogates

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Route Decomposition Prompt with preserved invariant and bridge theorem
- Route Decomposition Prompt correspondence theorem transport invariant
- Route Decomposition Prompt framework High-pressure obstruction classes are carrying multiple inferential roles at once, so route decomposition is needed before more local search. theorem
- theorem or framework matching: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.

Required Questions

- Which inferential roles are currently being conflated in one bridge object?
- What is the smallest theorem that isolates each role?
- Which decomposed segment is most likely to produce a standalone contribution?

Search Modes

- activation theorem search
- bridge split
- bridge-localization refinement
- counterfactual sharpening
- cross-model transport principle
- equivalence search between live carrier encodings
- explicit carrier construction
- hypothesis verification search
- invariant-preserving compilation theorem
- representation normal form for formula-size transfer
- surrogate problem design with forcing semantics
- typed abstraction search

13 disclayer.p-vs-np.family.barrier-aware-transfer

Kind. breakthrough-family
Target. barrier-aware transfer
Routing priority. 2.309
Dispatch rank. 10

Spawn reason. Spawn a dedicated 'barrier-aware transfer' discovery layer because obstruction 'local-reduction-barrier' carries repeated pressure across the blocked frontier.

Rationale. Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions.

System Prompt

You are searching under an explicit barrier theorem. Either construct a transfer that survives the barrier's constraints or sharpen the impossibility theorem. Do not use generic hardness language that ignores the reduction regime.

User Task

Describe a barrier-aware gadget family or impossibility result for the blocked transfer, with locality bounds and preserved witness structure made explicit.

Expected Deliverables

- State one theorem-shaped candidate centered on 'barrier-aware transfer' satisfying 'Either construct a local reduction family G satisfying locality bounds L and preserving witness structure W, or prove that any such G would contradict barrier theorem B.'.
- Make 'A barrier-aware reduction gadget family or an impossibility theorem explaining why the reduction regime cannot be crossed.' explicit enough that an expert could inspect the object or transfer law.
- Record one decisive negative route explaining what would fail if this discovery layer does not land.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'ignoring the known local-reduction barrier theorem'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'barrier-aware transfer' remains purely schematic after the first explicit object or theorem candidate.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. cross-family-pressure, bridge-family-foundation-gap

Candidate framework regions. lower-bound barrier frameworks, local-reduction impossibility theory, gadget and locality-preservation frameworks

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Barrier-Aware Transfer Prompt with preserved invariant and bridge theorem
- Barrier-Aware Transfer Prompt correspondence theorem transport invariant
- Barrier-Aware Transfer Prompt framework Design a transfer mechanism that survives known local-reduction barriers, or prove that no such mechanism exists under explicit assumptions. theorem
- theorem or framework matching: Either construct a local reduction family G satisfying locality bounds L and preserving witness structure W, or prove that any such G would contradict barrier theorem B.

Required Questions

- What exact barrier theorem constrains the route?
- What locality or regime parameter must any positive construction respect?
- What negative theorem would be a contribution if positive transfer fails?

Search Modes

- barrier-aware gadget design
- impossibility theorem search
- reduction regime decomposition

14 disclayer.p-vs-np.pattern.problem-representation-shift

Kind. epistemic-pattern

Target. problem-representation-shift

Routing priority. 6.548

Dispatch rank. 11

Spawn reason. Spawn 'problem-representation-shift' as a cross-cutting discovery layer because the blocked frontier spans multiple breakthrough families and needs a broader reframing.

Rationale. Several blocked routes look misrepresented rather than simply false, so representation change is itself a discovery pattern.

System Prompt

You are testing whether the blocked route is misrepresented rather than false. Search for an alternative encoding, normal form, dual object, or reformulation in which the route becomes mathematically transparent.

User Task

Propose representation changes that preserve the source-side content but expose a better target interface, and state the first theorem each representation would make possible.

Expected Deliverables

- Produce one reframing under 'problem-representation-shift' that changes the search state rather than restating the blocked lemma.
- Connect the reframing back to a theorem-shaped objective compatible with 'There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.'.
- Record one stop condition showing when the pattern has become superficial rather than discovery-bearing.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'adding a stronger carrier without specifying the target object class map'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'problem-representation-shift' only re-labels the blocked frontier instead of reducing proof debt or changing the representation.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. structural-analogy, cross-family-pressure, pattern-level-reframing, bridge-family-foundation-gap

Candidate framework regions. normal-form frameworks, equivalent reformulation theorems, model/encoding correspondence theories, proof complexity, pseudorandomness / derandomization

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Problem Representation Shift Prompt with preserved invariant and bridge theorem
- Problem Representation Shift Prompt correspondence theorem transport invariant
- Problem Representation Shift Prompt framework Several blocked routes look misrepresented rather than simply false, so representation change is itself a discovery pattern. theorem
- theorem or framework matching: There exists an admissible MCSP witness family, realized as a family of instances equipped with a monotone witness measure, such that whenever A barely-nontrivial lower bound for an explicit problem Q in a weak model holds for MCSP in nonuniform-circuits, the object organizes the source-side mathematical content while preserving the invariant MCSP complexity profile in nonuniform-circuits and yields major consequences such as NP not in P/poly.

Required Questions

- Which part of the current representation hides the real inferential structure?
- What alternative object language or normal form would expose the missing bridge?
- What mathematical content must stay invariant under the representation change?

Search Modes

- bridge split
- counterfactual sharpening
- cross-model transport principle
- dual characterization of the target object
- duality or normal form search
- generalization by faithful embedding
- invariant-preserving compilation theorem
- normal-form mediation
- object-class correspondence search
- parameter-preserving reformulation
- problem-specific to generic theorem bridge

- representation normal form for formula-size transfer
- semantic correspondence theorem
- two-way implication decomposition
- typed abstraction search

15 disclayer.p-vs-np.family.alignment

Kind. breakthrough-family

Target. alignment

Routing priority. 2.222

Dispatch rank. 12

Spawn reason. Spawn a dedicated 'alignment' discovery layer because obstruction 'target-alignment' carries repeated pressure across the blocked frontier.

Rationale. Identify or prove an equivalence between the current carrier class and the object class required by the target theorem.

System Prompt

You are aligning a local carrier class with the object class demanded by an existing target theorem. Topical similarity is not enough. Search for equivalence, correspondence, or a mediating normal form.

User Task

Describe the object-class map from the current carrier to the target theorem's object class, including the quantitative property that must survive the correspondence.

Expected Deliverables

- State one theorem-shaped candidate centered on 'alignment' satisfying 'Every admissible carrier C from the local route induces an object D in the target theorem's class, and D preserves the quantitative property needed to activate the target theorem.'.
- Make 'An alignment theorem, correspondence theorem, or mediating normal form linking the local carrier to the target's sparse-problem or distinguisher objects.' explicit enough that an expert could inspect the object or transfer law.
- Record one decisive negative route explaining what would fail if this discovery layer does not land.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'treating topical similarity as object equivalence'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.

- Escalate if 'alignment' remains purely schematic after the first explicit object or theorem candidate.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. structural-analogy, cross-family-pressure, bridge-family-foundation-gap

Candidate framework regions. sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Alignment Theorem Prompt with preserved invariant and bridge theorem
- Alignment Theorem Prompt correspondence theorem transport invariant
- Alignment Theorem Prompt framework Identify or prove an equivalence between the current carrier class and the object class required by the target theorem. theorem
- theorem or framework matching: Every admissible carrier C from the local route induces an object D in the target theorem's class, and D preserves the quantitative property needed to activate the target theorem.

Required Questions

- What is the target theorem's exact object class?
- How does the local carrier induce an object in that class?
- Which parameter or sparsity threshold is hardest to preserve?

Search Modes

- object-class correspondence search
- normal-form mediation
- dual characterization of the target object

16 disclayer.p-vs-np.family.explicitness

Kind. breakthrough-family

Target. explicitness

Routing priority. 2.107

Dispatch rank. 13

Spawn reason. Spawn a dedicated 'explicitness' discovery layer because obstruction 'pit-upgrade' carries repeated pressure across the blocked frontier.

Rationale. Upgrade a semantic or existential algebraic object into an explicit algorithmic derandomization premise.

System Prompt

You are upgrading a semantic or existential object into an explicit algorithmic one. Do not accept existence without uniformity, efficiency, or extractability.

User Task

Search for an explicit generator, certificate family, extractor, or uniform construction rule that turns the blocked semantic object into a derandomization-ready or algorithmically inspectable premise.

Expected Deliverables

- State one theorem-shaped candidate centered on 'explicitness' satisfying 'If semantic object C exists for the relevant circuit class, then one can uniformly and efficiently extract an explicit structure G that yields deterministic PIT for that class.'.
- Make 'An explicit generator, certificate family, or algebraic witness system that is constructive enough for PIT rather than merely semantic.' explicit enough that an expert could inspect the object or transfer law.
- Record one decisive negative route explaining what would fail if this discovery layer does not land.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'equating a black-box hitting-set statement with an explicit generator'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'explicitness' remains purely schematic after the first explicit object or theorem candidate.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. structural-analogy, cross-family-pressure, bridge-family-foundation-gap

Candidate framework regions. derandomization and generator frameworks, algebraic certificate constructions, extractor / hitter frameworks

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Explicitness Upgrade Prompt with preserved invariant and bridge theorem
- Explicitness Upgrade Prompt correspondence theorem transport invariant
- Explicitness Upgrade Prompt framework Upgrade a semantic or existential algebraic object into an explicit algorithmic derandomization premise. theorem
- theorem or framework matching: If semantic object C exists for the relevant circuit class, then one can uniformly and efficiently extract an explicit structure G that yields deterministic PIT for that class.

Required Questions

- What part of the current object is only existential or semantic?
- What explicit structure would replace that existential step?
- What efficiency class is required for the upgraded object to matter?

Search Modes

- explicit extractor or generator search
- certificate map construction
- semantic-to-algorithmic upgrade theorem

17 disclayer.p-vs-np.family.constructive-instantiation

Kind. breakthrough-family

Target. constructive instantiation

Routing priority. 1.966

Dispatch rank. 14

Spawn reason. Spawn a dedicated 'constructive instantiation' discovery layer because obstruction 'schema-to-examples' carries repeated pressure across the blocked frontier.

Rationale. Turn an abstract schema into explicit natural example families or prove that the schema cannot instantiate naturally in the current way.

System Prompt

You are turning an abstract schema into explicit mathematical examples. Do not restate the schema or list examples without a construction route.

User Task

Search for a construction template, reduction family, or canonicalization principle that produces explicit natural instances from the abstract schema.

Expected Deliverables

- State one theorem-shaped candidate centered on 'constructive instantiation' satisfying 'From abstract schema S, construct an explicit family of natural instances E_n satisfying properties P and realizing target theorem T.'.
- Make 'A construction template, example-generating reduction family, or a canonicalization principle for natural examples.' explicit enough that an expert could inspect the object or transfer law.
- Record one decisive negative route explaining what would fail if this discovery layer does not land.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'restating the schema as if it already supplied examples'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'constructive instantiation' remains purely schematic after the first explicit object or theorem candidate.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. cross-family-pressure

Candidate framework regions. canonical example construction frameworks, natural proof or canonicalization methods, explicit reduction template families

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Constructive Instantiation Prompt with preserved invariant and bridge theorem
- Constructive Instantiation Prompt correspondence theorem transport invariant
- Constructive Instantiation Prompt framework Turn an abstract schema into explicit natural example families or prove that the schema cannot instantiate naturally in the current way. theorem
- theorem or framework matching: From abstract schema S, construct an explicit family of natural instances E_n satisfying properties P and realizing target theorem T.

Required Questions

- What is the first explicit instance family that should be produced?
- What reduction or construction template yields that family?
- If the construction fails, what does that say about the schema's naturality?

Search Modes

- canonical example construction
- template instantiation
- example-preserving reduction search

18 disclayer.p-vs-np.family.semantic-bridge

Kind. breakthrough-family

Target. semantic bridge

Routing priority. 1.946

Dispatch rank. 15

Spawn reason. Spawn a dedicated 'semantic bridge' discovery layer because obstruction 'characterization-bridge' carries repeated pressure across the blocked frontier.

Rationale. Bridge a local definitional object to a global characterization theorem.

System Prompt

You are bridging a local definitional object to a global characterization theorem. Syntactic normalization is insufficient. Search for the missing semantic equivalence, duality, or witness-extraction principle.

User Task

Decompose the biconditional or characterization gap into the minimal semantic implications that would connect the local definition to the global theorem.

Expected Deliverables

- State one theorem-shaped candidate centered on 'semantic bridge' satisfying 'Local definitional object D has property P if and only if global characterization condition Q holds, with both directions proved at the right level of abstraction.'.
- Make 'A semantic correspondence, duality, witness extraction principle, or completeness notion that turns the local definition into the target biconditional.' explicit enough that an expert could inspect the object or transfer law.
- Record one decisive negative route explaining what would fail if this discovery layer does not land.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'normalizing syntax without adding semantic force'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'semantic bridge' remains purely schematic after the first explicit object or theorem candidate.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. structural-analogy, cross-family-pressure, bridge-family-foundation-gap

Candidate framework regions. duality frameworks, characterization theorems in proof complexity, semantic completeness correspondences

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Semantic Bridge Prompt with preserved invariant and bridge theorem
- Semantic Bridge Prompt correspondence theorem transport invariant
- Semantic Bridge Prompt framework Bridge a local definitional object to a global characterization theorem. theorem
- theorem or framework matching: Local definitional object D has property P if and only if global characterization condition Q holds, with both directions proved at the right level of abstraction.

Required Questions

- What semantic content is missing from the current definition-level object?
- Which direction of the characterization is harder and why?
- What duality or witness principle could mediate the bridge?

Search Modes

- semantic correspondence theorem
- duality or normal form search
- two-way implication decomposition

19 disclayer.p-vs-np.family.categorical-reframing

Kind. breakthrough-family

Target. categorical-reframing

Routing priority. 0.910

Dispatch rank. 16

Spawn reason. Spawn a dedicated 'categorical-reframing' discovery layer because obstruction 'structural-composition-gap' carries repeated pressure across the blocked frontier.

Rationale. Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation.

System Prompt

Use category theory as a structural and compositional discovery lens, not as decorative abstraction. Identify the source object, target object, typed morphisms already available, and the diagram that should commute if the route is real. Search for a mediating object with a universal property, test whether the route factors through it, and classify failure as no morphism, non-functorial transport, broken naturality, missing factorization, or failed universal property.

User Task

Reframe the blocked frontier as a family of diagrammatic composition problems. State candidate objects, morphisms, commutative squares, factorization claims, and universal properties that would make the live routes structurally compose.

Expected Deliverables

- State one theorem-shaped candidate centered on 'categorical-reframing' satisfying 'There exists a mediating object M and typed morphisms making the route commute; otherwise the failure is certified as no morphism, non-functorial transport, broken naturality, or missing universal property.'.
- Make 'A structural account of the route in terms of source and target objects, typed morphisms, a mediating universal object, and the factorization or naturality law that would make the route compose.' explicit enough that an expert could inspect the object or transfer law.
- Record one decisive negative route explaining what would fail if this discovery layer does not land.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'renaming route components as categorical objects without reducing proof debt'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'categorical-reframing' remains purely schematic after the first explicit object or theorem candidate.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. cross-family-pressure, bridge-family-foundation-gap

Candidate framework regions. sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, lower-bound barrier frameworks, local-reduction impossibility theory

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.
- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Categorical Reframing Prompt with preserved invariant and bridge theorem

- Categorical Reframing Prompt correspondence theorem transport invariant
- Categorical Reframing Prompt framework Recast the blocked frontier as a diagrammatic composition problem: identify objects, morphisms, mediating objects, and the exact commutativity or factorization obligation. theorem
- theorem or framework matching: There exists a mediating object M and typed morphisms making the route commute; otherwise the failure is certified as no morphism, non-functorial transport, broken naturality, or missing universal property.

Required Questions

- What are the source object, target object, and typed morphisms already present in the route?
- What diagram should commute if the route is legitimate?
- What mediating object would be universal for this route?
- Does the route fail because there is no morphism, because the square does not commute, or because the factorization is missing?

Search Modes

- diagram construction
- commutativity checking
- factorization search
- universal-property search
- adjunction-style dualization

20 disclayer.p-vs-np.family.equivalence-coherence-reframing

Kind. breakthrough-family

Target. equivalence-coherence-reframing

Routing priority. 0.880

Dispatch rank. 17

Spawn reason. Spawn a dedicated 'equivalence-coherence-reframing' discovery layer because obstruction 'equivalence-coherence-gap' carries repeated pressure across the blocked frontier.

Rationale. Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge.

System Prompt

Use HoTT as an equivalence and coherence discovery lens, not as a replacement foundation. Type the active concepts and bridge objects, ask whether two formulations are equal, equivalent, or only analogous, search for transportable invariants, and classify failure as no equivalence witness, incoherent transport, or weak equivalence where strict identity is needed.

User Task

Reframe the blocked frontier as a problem about sameness, equivalence, transport, and coherence. State the candidate equivalence claims, the transport obligations they would induce, and the coherence failures that currently block the route.

Expected Deliverables

- State one theorem-shaped candidate centered on 'equivalence-coherence-reframing' satisfying 'Either exhibit an equivalence E between the live formulations and prove that the target-relevant invariant transports coherently along E, or prove that no such equivalence witness exists in the current regime.'.
- Make 'A typed account of active formulations as types and terms, an explicit equivalence witness or non-equivalence diagnosis, and the transport law needed to move invariants and theorem hypotheses across that equivalence.' explicit enough that an expert could inspect the object or transfer law.
- Record one decisive negative route explaining what would fail if this discovery layer does not land.

Stop Criteria

- Stop immediately if the branch collapses into the anti-pattern 'treating topical similarity as equivalence'.
- Stop if the branch yields no theorem-shaped claim, carrier, transport principle, invariant, or representation change.
- Stop after one explicit positive candidate and one explicit failure probe have both been recorded.
- Escalate if 'equivalence-coherence-reframing' remains purely schematic after the first explicit object or theorem candidate.

External Framework Search Heuristic

Recommended. True

Mode. targeted-foundation-search

Justification. Search externally for a mathematical framework with similar structural properties only after local routes, MathLib, and explicit negative probes fail to provide the missing carrier, invariant, transfer law, or correspondence theorem.

Triggers. structural-analogy, cross-family-pressure, bridge-family-foundation-gap

Candidate framework regions. sparse-object and distinguisher frameworks, duality or correspondence theorems, representation-theoretic normal forms, derandomization and generator frameworks, algebraic certificate constructions

Local Prerequisites Before Browsing

- Record the failed local theorem lookup, MathLib lookup, and negative probe that justify leaving local search.
- State the missing operation concretely: carrier realization, transport, alignment, explicitness, characterization, or invariant discovery.

- Reject at least one purely topical analogy before searching for an external framework.

Query Templates

- mathematical framework for Equivalence / Coherence Reframing Prompt with preserved invariant and bridge theorem
- Equivalence / Coherence Reframing Prompt correspondence theorem transport invariant
- Equivalence / Coherence Reframing Prompt framework Recast the blocked frontier as an identity, equivalence, transport, or coherence problem rather than as a missing brute-force bridge. theorem
- theorem or framework matching: Either exhibit an equivalence E between the live formulations and prove that the target-relevant invariant transports coherently along E, or prove that no such equivalence witness exists in the current regime.

Required Questions

- Which active concepts should be treated as types, and which concrete witnesses should be treated as terms?
- Are the live operationalizations equal, equivalent, incomparable, or only analogous?
- What invariant or theorem hypothesis must transport along the proposed equivalence?
- Where does coherence fail when the same route is identified in two different ways?

Search Modes

- equivalence witness search
- transport-law search
- coherence checking
- identity-vs-equivalence discrimination